CSE105/209A: Modern Algorithmic Toolbox,[*] PSET-2

Ian Chi 2253175

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1.
$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{1}$$

subtract λ from diagonal

$$A - \lambda I = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} \tag{2}$$

the characteristic polynomial is

$$\lambda(\lambda(\lambda^2-1)-\lambda)+(-1(\lambda^2-1)) = \lambda^4-3\lambda^2+1 = (\lambda^2-\lambda-1)(\lambda^2+\lambda-1) \tag{3}$$

which gives roots of

$$\begin{aligned} \lambda_1 &= -\frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_2 &= \frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_3 &= -\frac{1}{2} + \frac{\sqrt{5}}{2} \\ \lambda_4 &= \frac{1}{2} + \frac{\sqrt{5}}{2} \end{aligned} \tag{4}$$

[*] PI: Vaggos Chatziafratis

to find eigenvalues, find the null space after subtracting λI from the diagonal of A.

$$(A - \lambda I)x = 0 = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} x = 0 \tag{5}$$

$$\begin{aligned} -\lambda x_1 + x_2 &= 0 \\ x_1 - \lambda x_2 + x_3 &= 0 \\ x_2 - \lambda x_3 + x_4 &= 0 \\ x_3 - \lambda x_4 &= 0 \end{aligned} \tag{6}$$

if we set x_1 to 1,

$$\begin{aligned} x_2 &= \lambda \\ x_3 &= \lambda^2 - 1 = (\lambda - 1)(\lambda + 1) \\ x_4 &= \lambda - 1/\lambda \end{aligned} \tag{7}$$

note

$$[\frac{1}{2}(\pm 1 \pm \sqrt{5}) - 1][\frac{1}{2}(\pm 1 \pm \sqrt{5}) + 1] = \frac{1}{2}(\mp 1 \mp \sqrt{5}) - 1 \tag{8}$$

for λ_1,

$$v_1 = \left(1, -\frac{1}{2} - \frac{\sqrt{5}}{2}, \frac{1}{2} + \frac{\sqrt{5}}{2}, \right) \tag{9}$$

2.

$$B = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \tag{10}$$

$$B - \lambda I = \begin{pmatrix} 1-\lambda & -1 & 0 & 0 \\ -1 & 2-\lambda & -1 & 0 \\ 0 & -1 & 2-\lambda & -1 \\ 0 & 0 & -1 & 1-\lambda \end{pmatrix} \tag{11}$$

the characteristic polynomial of B is